

Beyond Pass@1: Reliability and Token-Cost Effects of Quantized Reasoning Models under a Pinned Serving Stack

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Abstract

Post-training quantization is a common serving default for reasoning language models, but pass@1 on a single seed can hide changes in trace length, termination behavior, and sample-level reliability. We pin one serving stack—vLLM 0.7.0 [10] eager execution on an NVIDIA A100-80GB, with FP8 checkpoints executed as Marlin W8A16 [5] rather than native W8A8—and vary only the weight checkpoint among BF16, FP8, AWQ-4, and GPTQ-4 for DeepSeek-R1-Distill-Qwen-7B and DeepSeek-R1-Distill-Llama-8B. The grid has 88 cells and 56,408 completions on MATH-500 [8, 13] (5 seeds), GSM8K [2] (3 seeds), and GPQA-Diamond [16] (3 seeds).

On MATH-500, FP8-BF16 pass@1 differences are +0.40 and +0.28 percentage points (pp). Problem-clustered bootstrap 95% intervals include zero; a ± 1 pp TOST equivalence test is *not* passed. Four-bit effects are architecture- and task-dependent: Llama AWQ-4 loses 2.76 pp on MATH-500 (95% CI $[-4.16, -1.44]$, $p < 0.001$) and 1.57 pp on GSM8K; Qwen AWQ-4 loses 5.56 pp on GPQA-Diamond (95% CI $[-9.60, -1.52]$, $p = 0.007$). Majority-vote McNemar tests on maj@5 remain non-significant and do not answer the pass@1 question. Mean MATH-500 length rises 6.3–6.8% for Qwen 4-bit (paired token CIs exclude zero); the increase is concentrated on items where BF16 is correct and the quantized run is not. Identical-word loops are rare but nonzero (25/56,408); none of the rows hit the 32,768 token cap after re-encoding, yet 209 completions sit at $\geq 32,500$ tokens, and Qwen 4-bit near-cap counts on MATH-500 are about $1.7\times$ BF16. Using recovered answer strings, gold-free modal agreement is an observable abstention signal: 5/5 consensus has very low selective error on MATH-500 but costs five generations per query, and Llama AWQ-4 reduces 5/5 coverage by 6.0 pp versus BF16 (95% paired CI $[-9.4, -2.6]$). On a controlled serving confirmation (balanced MATH-500 subsets, `max_num_seqs=8` pinned, same-node within architecture), batched Qwen-7B GPTQ-4 has the lowest scenario Cost-of-Pass among Qwen cells (−45.9% vs. BF16 at \$1.50/A100-hour); Qwen FP8 is −36.0% C_{pass} at +78% mean tok/s, with a large wall-clock spread (449.8 ± 97.5 tok/s; identical token counts). Single-request serving effects were architecture dependent: Qwen AWQ/GPTQ substantially exceeded BF16 throughput, whereas Llama AWQ/GPTQ did not. Peak allocated VRAM stays ~ 54 –56 GB across formats under `gpu_memory_utilization=0.75` and does not measure weight-footprint savings.

Keywords: Reasoning language models, post-training quantization, pinned serving stack, token inflation, selective prediction, Cost-of-Pass.

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1 Introduction

Reasoning models such as DeepSeek-R1 [7] emit long traces before a final answer. Weight-only post-training quantization (FP8, AWQ-4 [14], GPTQ-4 [4]) is therefore a common serving choice. Isolated pass@1 numbers hide whether compression also changes trace length [12, 15], termination, or the cost of a correct answer [3]. Serving-stack details—engine version, decoding flags, CUDA graphs—can move reasoning scores as much as model changes [9]. This paper *pins* one stack rather than shifting it: we do not run a factorial vLLM 0.7.0 vs. 0.8.5 experiment.

We ask four questions under that pin:

- **RQ1.** Does quantization change *pass@1* relative to matched BF16, with problem-clustered uncertainty?
- **RQ2.** Do identical-word loops and near-budget terminations occur, and do they differ by format?
- **RQ3.** What can be said about multi-sample reliability without gold labels at serve time?
- **RQ4.** How do measured decode throughput and GPU-seconds per query change Cost-of-Pass relative to a shared-throughput token proxy?

Contributions. (1) A released, format-matched, stack-pinned 88-cell grid (56,408 completions) with per-cell JSON. (2) Problem-clustered pass@1 inference showing FP8–BF16 intervals that include zero, a statistically detectable Llama AWQ-4 drop on MATH-500/GSM8K, and a Qwen AWQ-4 drop on GPQA-Diamond—effects that maj@5 McNemar tests miss. (3) Evidence that 4-bit token inflation is real for Qwen on the full MATH-500 grid, is *not* an artifact of a 200-item even-index subset, and is concentrated on format-induced failures rather than uniformly longer correct traces. (4) Gold-free modal-answer agreement on MATH-500: higher observable agreement is associated with lower selective error, while 4-bit AWQ reduces strict-consensus coverage in some configurations even when selective error among retained predictions stays low. (5) Controlled measured serving on the same pinned stack (52 task-realistic confirmation runs + 8 fixed-token microbenchmarks; `max_num_seqs=8`): scenario Cost-of-Pass uses campaign MATH-500 pass@1 and measured GPU-seconds rather than a shared 65 tok/s assumption. An earlier unconstrained serving timing is retained only as provenance.

2 Related Work

GPTQ [4] and AWQ [14] were originally validated mainly on short-form benchmarks; SmoothQuant [18] addresses activation quantization, which we do not run. Liu et al. [15] evaluate DeepSeek-R1 distillations across weight, activation, and KV quantization and report near-lossless W8/W4A16 in many settings, with *no* systematic output lengthening. Lian et al. [12] report the opposite: low-bit PTQ can inflate chain-of-thought length. Li et al. [11] localize 4-bit math failures to early process errors. Alimaskina et al. [1] study loops and delayed commitment under extreme low-bit inference. Our delta is a single pinned vLLM 0.7.0 eager path [10] with five MATH-500 seeds and paired item-level tests, not a new compressor.

Hochlehnert et al. [9] show that seeds, prompts, decoding, and software move reasoning scores; we freeze those factors. Zhong et al. [19] study uncertainty calibration of quantized LLMs using model probabilities—a quantity we do not log. Self-consistency [17] and later unsupervised estimators [20] treat sample agreement as confidence; selective classification [6] supplies the risk–coverage language. Erol et al. [3] define Cost-of-Pass as expected monetary cost per correct solution; we report it twice: a historical shared-65 tok/s token proxy, and a measured GPU-second version on a frozen MATH-500 serving subset.

3 Experimental Methodology

3.1 Models and precision formats

We evaluate DeepSeek-R1-Distill-Qwen-7B and DeepSeek-R1-Distill-Llama-8B [7]. Formats: uncompressed **BF16**; **FP8** checkpoints served on A100 via vLLM’s Marlin **W8A16** weight-only fallback (not native W8A8 tensor-core FP8); **AWQ-4** (group size 128, float16); **GPTQ-4** (group size 128, Marlin). Hugging Face IDs and revisions are listed in the appendix. AWQ-4 checkpoints are third-party community uploads; “Llama AWQ-4 is weak” is a property of those artifacts, not a proof about AWQ as a method.

3.2 Benchmarks and protocol

MATH-500 [8, 13] ($n = 500$, seeds 42–46): 20,000 completions. GSM8K [2] ($n = 1,319$, seeds 42–44) and GPQA-Diamond [16] ($n = 198$, seeds 42–44): breadth. Decoding: zero-shot <think> prefix (appendix), $T = 0.6$, top- $p = 0.95$, repetition penalty 1.0 (deliberately not used to hide loops), max new tokens 32,768. Total: 56,408 completions in 88 cells. MATH and GSM8K are widely used; contamination likely inflates absolute scores but largely cancels in within-model format contrasts.

3.3 Hardware and serving stack

PARAM Rudra HPC (IIT Patna / NSM), NVIDIA A100-PCIE-80GB. Publication runs use the official QRM inference.py path [15] with `gpu_memory_utilization=0.75`. Engine pin: `vLLM==0.7.0`, eager execution. Source files under `configs/models/` are *not* the campaign launcher; they differ in `max_model_len` and KV-cache defaults and should not be read as the effective stack.

3.4 Measured serving

After the 56k accuracy campaign we timed the same eight checkpoints on A100-80GB under vLLM 0.7.0 eager. Primary numbers are the *controlled confirmation* (protocol: `docs/MEASURED_SERVING_CONFIRMATION`, raw JSON under `results/measured_serving_confirmation/raw/`). Condition A uses a balanced 20-prompt MATH-500 subset (4 per difficulty level), sequential single-request decode. Condition B uses a balanced 100-prompt subset (20 per level) in one `llm.generate` call with `max_num_seqs=8` pinned. Secondary microbenchmark: 10 prompts forced to exactly 512 tokens (`ignore_eos`). Three warmup queries are discarded. Sampling matches the campaign ($T = 0.6$, top- $p = 0.95$) with seed 20260816. Repetitions share that seed, so output token counts are identical across reps; the spread is wall-clock noise, not generation variability. The runner starts at $R = 3$ and expands to $R = 5$ when the *population* CV of the first three tok/s values exceeds 3%; reported $\pm$ is *sample* SD. Qwen-7B jobs ran on `ragpu003` and Llama-8B jobs on `ragpu004` (one visible A100 each; PyTorch 2.5.1+cu124). Cross-architecture absolute tok/s is therefore cautious. An earlier unconstrained timing (`docs/MEASURED_SERVING_PROTOCOL.md`; mixed nodes; `max_num_seqs` not pinned) is not averaged with these numbers and is not used in the main tables.

Output tok/s = $(\sum \text{output tokens}) / \Delta t_{\text{steady}}$. GPU-sec/query = $\Delta t_{\text{steady}} / N$ on one GPU. Peak VRAM is `torch.cuda.max_memory_allocated` after the 0.75 utilization preallocation (~ 60 GB reserved of 80 GB). Measured

$$C_{\text{pass}}^{\text{meas}} = \frac{(\text{GPU-sec/query}) \times (\$1.50/3600)}{\text{pass@1}}, \quad (1)$$

using the frozen 40-cell MATH-500 means—not accuracy on the 100-item subset. The \$1.50/A100-hour rate is a cloud pricing *scenario*, not a billed cluster price.

3.5 Metrics

Pass@1. Mean extractive match across seeds. Primary inference is a problem-clustered bootstrap of the paired difference $\Delta = \overline{\text{acc}}_{\text{quant}} - \overline{\text{acc}}_{\text{BF16}}$ (resample n items with replacement; retain all seeds per item; $B = 10,000$). We report Δ in percentage points, a 95% percentile CI, and a two-sided bootstrap p -value, with Holm–Bonferroni across the six MATH (resp. GSM8K, GPQA) contrasts. A ± 1 pp two one-sided tests (TOST) using the 90% CI is reported as an equivalence check, not as a default claim.

maj@5 McNemar (secondary). A problem is maj@5-correct if at least 3 of 5 MATH-500 seeds match gold. Exact McNemar tests this consensus, not pass@1.

Pathology. A row is a *loop* if the longest consecutive identical-word run is ≥ 20 (Unicode word tokens, case-folded). A row is a *cap hit* if re-encoded completion length is $\geq 32,768$. Because decode-then-re-tokenize undershoots the cap (observed max 32,737), we also count *near-cap* rows with $\geq 32,500$ tokens. Saved rows do not include vLLM `finish_reason` or output token IDs; phrase-level cycles are invisible to the identical-word detector.

Length. Primary estimator: ratio of means over all seeds, plus the mean paired per-item token delta with a clustered bootstrap CI. We also report the mean of per-item ratios, which is a different estimand and is dominated by short BF16 traces. Length is stratified by the 2×2 of BF16/quantized extractive correctness.

Historical token-implied C_{pass} . Following Erol et al. [3], $C_{\text{pass}} = (\text{cost per query})/\text{pass@1}$ with

$$\text{Cost per query} = (\bar{T}/\tau) \times (R/3600), \quad (2)$$

$\tau = 65$ tok/s (unmeasured) and $R = \$1.50$ per A100-hour. Because τ and R are shared, ranking by C_{pass} is ranking by $\bar{T}/\text{pass@1}$. We retain this proxy only as a sensitivity to the pre-measurement draft; primary systems/cost evidence is the measured GPU-second version in Section 3.4 and Table 8.

Observable modal agreement. Compact per-cell JSON stores `extractive_match` but not extracted strings. After recovering the 20,000 MATH-500 answer strings, we form agreement *without* gold. Campaign extraction and clustering used LightEval 0.8.0 with math-verify (frozen in `docs/ANSWER_NORMALIZATION.md`). A later throwaway MacBook install of LightEval 0.8.1 was *not* used for canonical numbers. For item i and five generated predictions, let $\sim$ be that campaign mathematical-equivalence relation. Cluster the five predictions into connected components of $\sim$. Let m_i be the size of the unique largest class when one exists; otherwise the item is a non-unique-mode tie. Observable agreement is $a_i = m_i/5$. Serve at threshold $k/5$ only when a unique modal class satisfies $m_i \geq k$. The pattern (A, A, B, B, C) has $m_i = 2$ and is not served at $\geq 3/5$; (A, A, A, B, B) has unique mode A with $m_i = 3$ and is served at $\geq 3/5$. Gold is used only afterward to estimate selective accuracy and risk. All five samples are generated before the decision. The five-sample token-cost proxy T_5 is the sum of their completion tokens, including abstained queries. We report three discrete operating points, not a continuous risk–coverage curve. Gold-hit ECE (confidence = $c_i/5$, label = $\mathbb{I}(c_i \geq 3)$) remains a deterministic function of the gold-hit histogram and is omitted.

Table 1: MATH-500 headline grid ($n = 500$, seeds 42–46). Clustered 95% CIs resample problems. Loops: identical-word run ≥ 20 . Near-cap: completion tokens $\geq 32,500$. Exact cap hits after re-encoding: 0 in all cells.

Cell	42	43	44	45	46	Mean $\pm$ Std	Clust. 95%	Tok.	L	NC
Qwen BF16	94.4	94.0	93.8	94.6	93.2	94.00 ± 0.55	[92.2, 95.6]	4,011.4	0	14
Qwen FP8	94.4	95.2	94.8	92.6	95.0	94.40 ± 1.05	[92.8, 96.0]	4,007.7	0	15
Qwen AWQ-4	92.4	92.8	93.2	93.0	94.2	93.12 ± 0.67	[91.4, 94.8]	4,265.3	1	25
Qwen GPTQ-4	93.8	92.6	93.4	94.6	93.0	93.48 ± 0.77	[91.7, 95.2]	4,287.2	0	24
Llama BF16	89.0	88.4	90.2	89.8	88.8	89.24 ± 0.74	[87.2, 91.2]	4,656.6	3	19
Llama FP8	89.0	89.6	88.6	89.2	91.2	89.52 ± 1.01	[87.4, 91.6]	4,550.8	3	10
Llama AWQ-4	84.4	84.8	89.2	87.4	86.6	86.48 ± 1.96	[84.2, 88.7]	4,736.5	3	12
Llama GPTQ-4	88.0	89.6	86.8	89.4	90.8	88.92 ± 1.55	[86.9, 90.9]	4,840.5	2	21

Table 2: Primary: problem-clustered bootstrap of MATH-500 pass@1 versus BF16 ($B = 10,000$). Secondary: exact McNemar on maj@5. Holm at $\alpha = 0.05$ over six contrasts. TOST ± 1 pp uses the 90% CI.

Contrast	Δ (pp)	95% CI	p	Holm	McNemar p
Llama AWQ-4	-2.76	[-4.16, -1.44]	< 0.001	sig.	0.296
Qwen AWQ-4	-0.88	[-1.88, +0.08]	0.069	n.s.	1.000
Qwen GPTQ-4	-0.52	[-1.40, +0.36]	0.245	n.s.	0.754
Qwen FP8	+0.40	[-0.48, +1.28]	0.383	n.s.	0.581
Llama GPTQ-4	-0.32	[-1.56, +0.96]	0.619	n.s.	1.000
Llama FP8	+0.28	[-0.92, +1.48]	0.647	n.s.	1.000

4 Results

4.1 MATH-500 pass@1

Table 1 and Figure 5 show per-seed pass@1. FP8 point estimates sit 0.28–0.40 pp above BF16. Llama AWQ-4 is the weak MATH cell (−2.76 pp).

Table 2 is the test that matches RQ1. Llama AWQ-4’s clustered interval excludes zero ($p < 0.001$; Holm-significant among the six MATH contrasts). Qwen 4-bit MATH drops are negative but not Holm-significant. No MATH contrast, including FP8, has its 90% CI inside $[-1, +1]$ pp, so we do *not* claim statistical equivalence at a 1 pp serving margin. Secondary maj@5 McNemar p -values are all > 0.29 : majority voting absorbs the per-seed flips that pass@1 still sees.

4.2 GSM8K and GPQA-Diamond

Table 5: FP8 stays within about one point of BF16 on all three tasks. The largest accuracy effect in the study is Qwen AWQ-4 on GPQA-Diamond (−5.56 pp, 95% CI [−9.60, −1.52], $p = 0.007$, Holm-significant among six GPQA contrasts; 3 seeds, $n = 198$). Llama AWQ-4 also drops on GSM8K (−1.57 pp, 95% CI [−2.55, −0.58], $p = 0.002$). Other GSM8K Qwen contrasts pass TOST at ± 1 pp. GPQA intervals are wide; we treat non-AWQ GPQA contrasts as exploratory.

4.3 Loops and near-cap termination

Across 88 cells the validator records **25** repetition-flagged rows (0.044% of 56,408) and **0** exact cap hits. Earlier drafts reported 0/0 because analysis scripts read `truncation_count` / `repetition_flag_count` while the validator writes `token_limit_hits` / `repetition_rows`. Loops occur in BF16 as well as quantized cells (Llama MATH: 3/3/3/2 for BF16/FP8/AWQ/GPTQ).

Table 3: MATH-500 length versus BF16, all five seeds. RoM: ratio of means. Δ : mean paired per-item token difference with clustered 95% CI. “Both OK” / “BF16 only”: mean Δ on (item, seed) pairs with that correctness pattern.

Contrast	RoM	Mean Δ [95% CI]	Both OK	BF16 only
Qwen FP8	−0.1%	−4 [−142, +134]	+51	+5444
Qwen AWQ-4	+6.3%	+254 [+116, +398]	+166	+6915
Qwen GPTQ-4	+6.9%	+276 [+121, +431]	+136	+6892
Llama FP8	−2.3%	−106 [−284, +65]	+38	+2629
Llama AWQ-4	+1.7%	+80 [−109, +268]	+91	+3090
Llama GPTQ-4	+3.9%	+184 [+12, +359]	+66	+5915

Table 4: Historical token-implied Cost-of-Pass on MATH-500 at \$1.50/A100-hour and a shared 65 tok/s. Not measured wall-clock. Ranking equals ranking by mean tokens / pass@1. Primary cost evidence is Table 8.

Cell	Mean tok.	Cost / Q	C_{pass}
Qwen BF16	4,011.4	\$0.02571	\$0.02736
Qwen FP8	4,007.7	\$0.02569	\$0.02721
Qwen AWQ-4	4,265.3	\$0.02734	\$0.02936
Qwen GPTQ-4	4,287.2	\$0.02748	\$0.02940
Llama BF16	4,656.6	\$0.02985	\$0.03345
Llama FP8	4,550.8	\$0.02917	\$0.03259
Llama AWQ-4	4,736.5	\$0.03036	\$0.03511
Llama GPTQ-4	4,840.5	\$0.03103	\$0.03490

Three runs exceed 10,000 identical consecutive words. Near-cap rows ($\geq 32,500$ tokens) total 209; on MATH-500, Qwen AWQ-4/GPTQ-4 have 25 and 24 versus BF16’s 14, and almost all near-cap rows are unboxed. Without `finish_reason` these are inferred length terminations, not proven `length` stops.

4.4 Token length

On the *full* MATH-500 grid and all five seeds (Table 3, Figure 3), Qwen AWQ-4/GPTQ-4 generate +6.33%/+6.88% more tokens than BF16 (mean paired Δ +254 and +276 tokens; CIs exclude zero). Llama GPTQ-4 is +3.95% (+184 tokens, CI just excludes zero). Llama AWQ-4 +1.72% and both FP8 cells have CIs that include zero.

A previous 200-item “stratified mixed-correctness” analysis used even indices among the first 400 problems, seed 42 only, and the *mean of per-item ratios*. That estimator is not the full-grid ratio of means: on the same 200 items it reports Qwen FP8 +11.4% while the ratio of means is −3.4%. We discard that subset interpretation.

Figure 4 and Table 3 split paired token deltas by correctness. When both formats are correct, 4-bit adds on the order of +90 to +170 tokens. When BF16 is correct and the quantized sample is wrong, the quantized trace is thousands of tokens longer. Token inflation is therefore largely a *failure-associated* lengthening, which is a more useful reconciliation of Liu et al. [15] (little mean lengthening) and Lian et al. [12] (hidden token cost) than a 200-item even-index slice.

Under the shared- τ model (Table 4), FP8 has the lowest token-implied C_{pass} by \$0.00015 on Qwen. That margin is smaller than seed noise in length. We do not call it Pareto-optimal. Measured batched decode on this stack is an order of magnitude faster than 65 tok/s, so the proxy overstates dollars; relative 4-bit rankings can also flip (Section 4.7).

Table 5: Cross-benchmark pass@1 (mean $\pm$ seed std). MATH-500: 5 seeds. GSM8K and GPQA-Diamond: 3 seeds. L/NC: loops / near-cap over those seeds.

Cell	MATH-500	GSM8K	GPQA-D	L / NC
Qwen BF16	94.00 $\pm$ 0.55	91.26 $\pm$ 0.29	50.34 $\pm$ 2.96	4 / 28
Qwen FP8	94.40 $\pm$ 1.05	91.33 $\pm$ 0.16	49.49 $\pm$ 1.52	0 / 24
Qwen AWQ-4	93.12 $\pm$ 0.67	91.05 $\pm$ 1.14	44.78 $\pm$ 3.04	5 / 32
Qwen GPTQ-4	93.48 $\pm$ 0.77	91.13 $\pm$ 0.27	47.98 $\pm$ 1.75	3 / 40
Llama BF16	89.24 $\pm$ 0.74	88.68 $\pm$ 0.46	46.13 $\pm$ 1.91	3 / 21
Llama FP8	89.52 $\pm$ 1.01	88.80 $\pm$ 0.62	47.81 $\pm$ 0.29	3 / 12
Llama AWQ-4	86.48 $\pm$ 1.96	87.11 $\pm$ 0.23	46.97 $\pm$ 2.02	4 / 21
Llama GPTQ-4	88.92 $\pm$ 1.55	88.96 $\pm$ 0.70	44.95 $\pm$ 4.32	3 / 31

Table 6: MATH-500 difficulty labels (5-seed pass@1 mean $\pm$ std). Labels are not monotone: Level 2 is harder than Level 5 for both BF16 backbones. Level-1 $n = 43$; treat cell-wise gaps as descriptive.

Level	Qwen BF16	Qwen FP8	Qwen AWQ-4	Qwen GPTQ-4
1 ($n = 43$)	98.60 $\pm$ 1.27	98.60 $\pm$ 2.08	98.14 $\pm$ 1.95	96.28 $\pm$ 2.08
2 ($n = 90$)	89.11 $\pm$ 0.93	90.44 $\pm$ 4.20	90.67 $\pm$ 1.69	88.67 $\pm$ 2.14
3 ($n = 105$)	95.81 $\pm$ 1.09	95.24 $\pm$ 1.51	94.10 $\pm$ 0.80	94.29 $\pm$ 1.90
4 ($n = 128$)	95.78 $\pm$ 0.89	96.41 $\pm$ 1.42	94.53 $\pm$ 1.24	96.56 $\pm$ 1.18
5 ($n = 134$)	92.69 $\pm$ 1.11	93.13 $\pm$ 1.23	91.04 $\pm$ 1.67	92.24 $\pm$ 1.95
Level	Llama BF16	Llama FP8	Llama AWQ-4	Llama GPTQ-4
1 ($n = 43$)	93.95 $\pm$ 3.12	94.88 $\pm$ 3.03	95.35 $\pm$ 1.64	96.28 $\pm$ 2.65
2 ($n = 90$)	84.67 $\pm$ 2.14	85.11 $\pm$ 2.02	81.11 $\pm$ 3.04	83.33 $\pm$ 3.42
3 ($n = 105$)	89.90 $\pm$ 2.48	90.10 $\pm$ 1.73	85.71 $\pm$ 1.17	89.71 $\pm$ 1.83
4 ($n = 128$)	91.56 $\pm$ 1.69	91.09 $\pm$ 0.43	87.03 $\pm$ 2.25	89.84 $\pm$ 1.75
5 ($n = 134$)	88.06 $\pm$ 1.75	88.81 $\pm$ 2.84	87.31 $\pm$ 3.46	88.81 $\pm$ 1.18

4.5 Difficulty labels

Table 6: Level 1 is easy for every format in this sample. Llama AWQ-4 is weakest on Levels 2–4. MATH level labels are not a monotone difficulty axis here (Level 2 Qwen BF16 89.11% vs Level 5 92.69%).

4.6 Observable modal agreement on MATH-500

Recovered answer strings make RQ3 measurable as observable multi-sample agreement, selective prediction, and abstention—not as model safety or softmax calibration. Table 7 reports coverage and selective risk at three discrete thresholds. On MATH-500, higher observable answer agreement is associated with lower selective error. At 5/5 consensus, selective error is at most 0.27% across the evaluated configurations, but coverage ranges from 70.2% (Llama AWQ-4) to 88.8% (Qwen FP8). Point estimates of 100% selective accuracy among served items are finite-sample statements and do not imply zero true error or guaranteed correctness.

AWQ-4 reduces strict-consensus coverage in some configurations, most clearly for Llama-8B: 70.2% versus 76.2% for BF16, a paired difference of -6.0 pp (95% CI $[-9.4, -2.6]$). Llama AWQ-4 $\geq 4/5$ coverage falls 3.2 pp ($[-6.0, -0.6]$). Qwen AWQ-4 $\geq 4/5$ coverage falls 2.2 pp ($[-4.0, -0.6]$). The $\geq 3/5$ AWQ contrasts, and all GPTQ-4 coverage contrasts, have paired intervals that include zero. No clear FP8–BF16 difference was detected in modal-agreement coverage or selective risk under these configurations. Appendix Table 9 lists the full threshold pattern.

Table 7: Observable modal-answer agreement on MATH-500 ($n = 500$, seeds 42–46). Coverage is the fraction of items with a unique modal class of size $\geq k$. Risk is 1–selective accuracy among served items. Mean T_5 is the five-sample token-cost proxy over all 500 items (abstained queries still pay). Gold is used only to score served predictions.

Model	Fmt	$\geq 3/5$ cov.	$\geq 3/5$ risk	$\geq 4/5$ cov.	$\geq 4/5$ risk	$5/5$ cov.	$5/5$ risk	Mean T_5
Qwen-7B	BF16	96.0	1.67	93.4	0.43	88.4	0.23	20,057
Qwen-7B	FP8	96.6	1.66	92.6	0.00	88.8	0.00	20,038
Qwen-7B	AWQ-4	95.8	1.46	91.2	0.44	86.4	0.23	21,327
Qwen-7B	GPTQ-4	95.4	1.47	92.8	0.43	86.8	0.00	21,436
Llama-8B	BF16	94.0	2.98	87.8	0.68	76.2	0.26	23,283
Llama-8B	FP8	94.0	3.19	88.6	0.90	78.2	0.00	22,754
Llama-8B	AWQ-4	93.2	3.65	84.6	1.89	70.2	0.00	23,682
Llama-8B	GPTQ-4	93.8	2.77	86.4	0.46	75.4	0.27	24,203

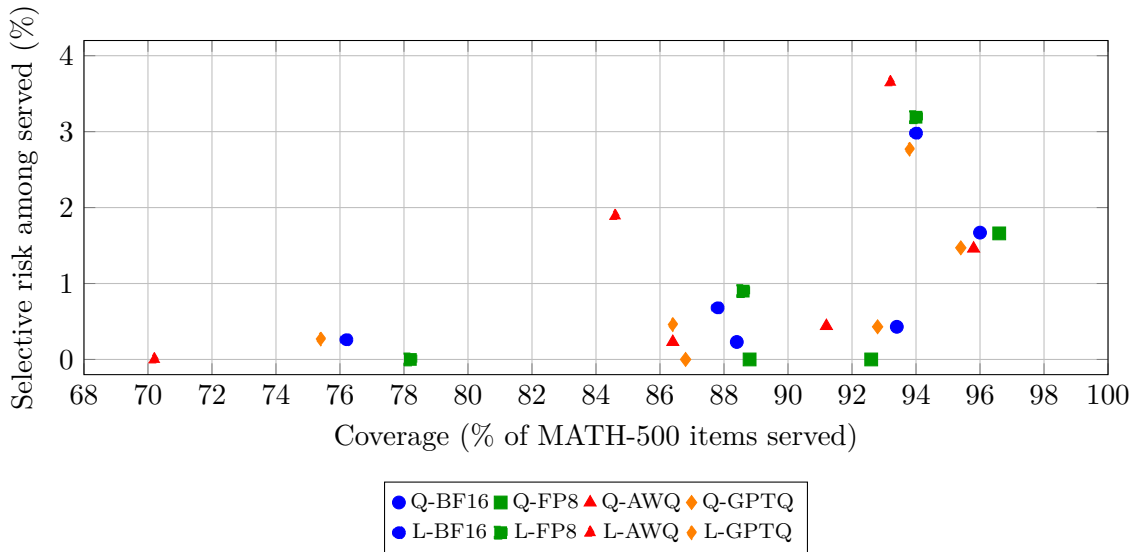


Figure 1: Three discrete MATH-500 operating points ($\geq 3/5$, $\geq 4/5$, $5/5$), not a continuous risk–coverage curve. Solid markers: Qwen-7B. Dashed markers: Llama-8B. Lower-right is high coverage and higher residual risk; lower-left is strict consensus.

The five-sample token-cost proxy is paid before abstention. Mean T_5 is 20,057 / 20,038 / 21,327 / 21,436 tokens for Qwen BF16/FP8/AWQ-4/GPTQ-4 and 23,283 / 22,754 / 23,682 / 24,203 for Llama. Qwen 4-bit retains its approximately +6–7% output-token penalty in this accounting because T_5 is the sum of the same five traces. Tokens per served query divide that total by the number served, so abstained items are not free. This is not a measured monetary cost. Figure 1 shows the three operating points.

4.7 Measured serving throughput and Cost-of-Pass

Table 8 reports the confirmation batched condition (100 balanced prompts, `max_num_seqs=8`). Qwen-7B GPTQ-4 is the lowest scenario C_{pass} among Qwen cells (−45.9% vs. BF16; \$0.0032 vs. \$0.0060) at \$1.50/A100-hour. Qwen FP8 is −36.0% C_{pass} and +78% mean tok/s (449.79 ± 97.53 vs. 252.72 ± 0.56), but that FP8 mean averages five wall-clock repeats with identical generated tokens and two distinct timing regimes (~ 351 vs. ~ 545 tok/s); we retain all five and do not drop the slow repeats. Llama-8B FP8 is +31.2% tok/s and −8.2% C_{pass} . Llama AWQ-4 and GPTQ-4 *raise* scenario C_{pass} (+18.5% and +24.9%) despite compression. These FP8 numbers

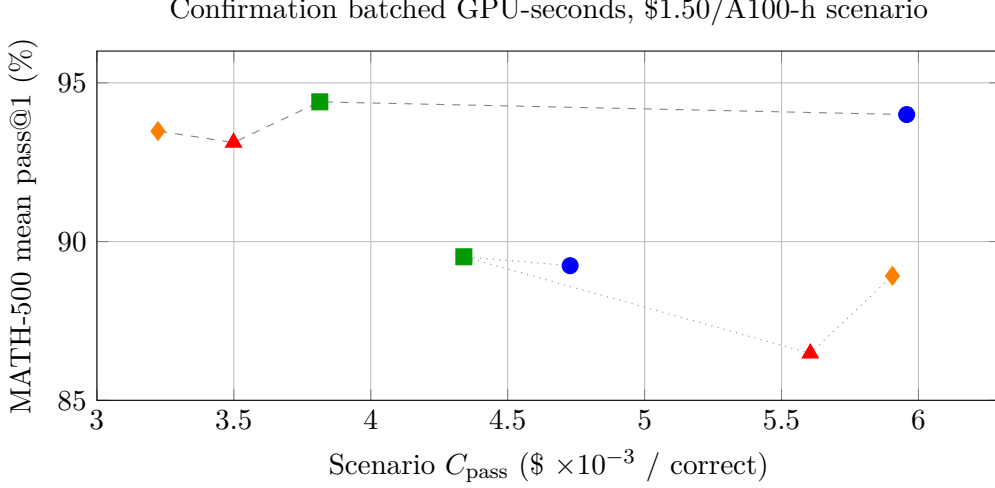


Figure 2: Confirmation Condition B scenario C_{pass} versus campaign MATH-500 pass@1 ($\$ \times 10^{-3}$). Solid/dashed lines are Qwen/Llama visual guides, not a fitted frontier. Pooled non-dominated on (pass@1, C_{pass}): Qwen-7B FP8 and Qwen-7B GPTQ-4. Within Llama: Llama-8B FP8 only. Not a unique “true Pareto optimum.”

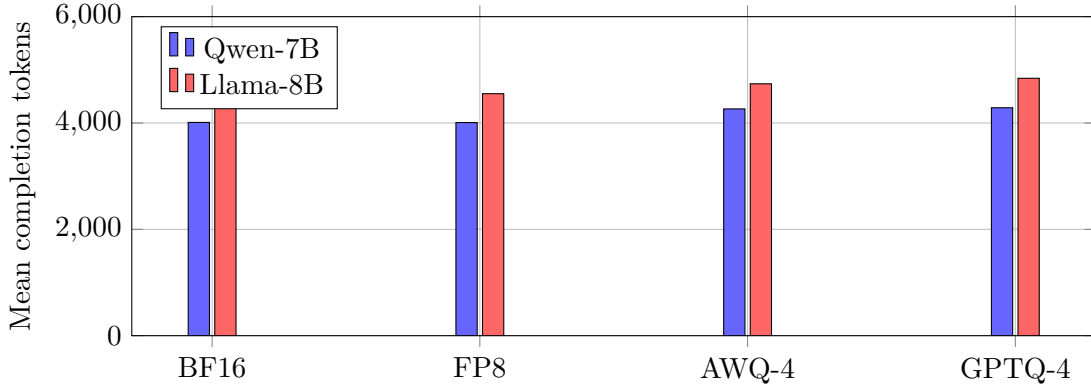


Figure 3: Full-grid mean MATH-500 completion tokens (ratio-of-means estimand).

are A100 Marlin W8A16, not Hopper W8A8.

Four-bit speed is not one number, and it is not uniformly a slowdown. Confirmation Condition A (Appendix Table 10): Qwen AWQ-4 and GPTQ-4 substantially exceed matched BF16 tok/s (76.89 and 82.67 vs. 43.92), whereas Llama AWQ-4 and GPTQ-4 do not (70.20 and 63.49 vs. 72.34). Batched Qwen GPTQ-4 reaches 488.26 ± 0.64 tok/s (+93.2% vs. Qwen BF16) under pinned concurrency 8; Llama GPTQ-4 is statistically tied with Llama BF16 (−0.2%). Serving effects are therefore architecture-, concurrency-, and implementation-dependent.

Token inflation does not automatically cancel a kernel speedup. Cost is GPU-seconds $\times$ a pricing scenario, not tokens alone. No serving-confirmation *mean* output reached the 32,768-token generation cap. Compact confirmation JSON does not store native `finish_reason`.

Peak allocated VRAM is 53.9–56.1 GB for every batched confirmation cell. That is the 0.75 utilization pool, not a measurement of 4-bit weight-footprint savings. We do not publish a format-to-format weight table from `torch.cuda.memory_allocated` after engine init. Memory and latency are separate.

On (pass@1, scenario C_{pass}) over the eight confirmation batched points, the pooled nondominated set is {Qwen-7B FP8, Qwen-7B GPTQ-4}. Within Qwen the same two points remain; AWQ-4 is dominated by GPTQ-4. Within Llama the only nondominated point is Llama FP8.

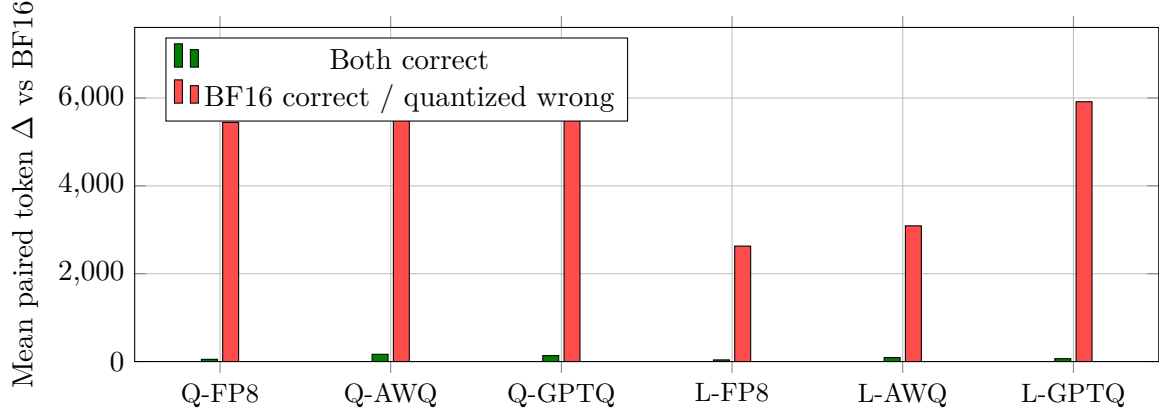


Figure 4: Paired token deltas by correctness. Extra length is concentrated on format-induced failures, not on uniformly longer correct traces.

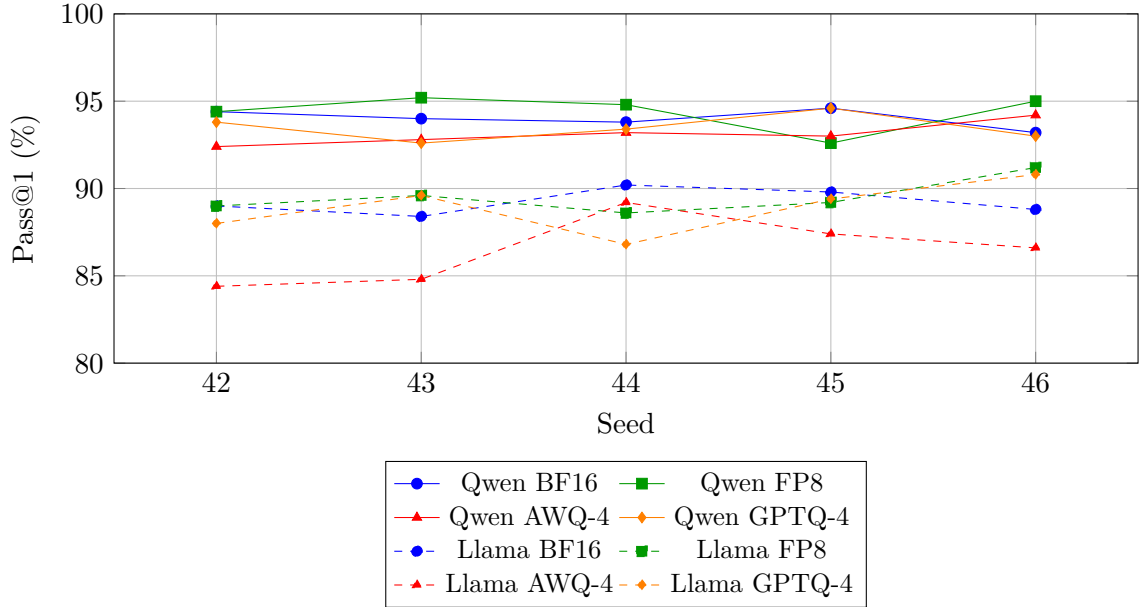


Figure 5: MATH-500 pass@1 by seed. Solid: Qwen-7B. Dashed: Llama-8B. Llama AWQ-4 is the unstable 4-bit cell.

We do not call any cell a “true Pareto optimum”: the frontier is two-objective, scenario-priced, and stack-specific (Figure 2). The old 65 tok/s proxy (Table 4) remains a sensitivity, not the primary cost evidence.

Table 8: Controlled confirmation batched serving (Condition B: 100-prompt `llm.generate`, `max_num_seqs=8` pinned). Output tok/s is mean $\pm$ sample SD. $R = 3$ except Qwen FP8 ($R = 5$; CV-triggered). GPU-sec/query = $\Delta t/N$ on one visible A100. C_{pass} uses campaign MATH-500 pass@1 and a \$1.50/A100-hour *scenario*. Peak VRAM is allocated bytes after `gpu_memory_utilization=0.75`, not isolated weight footprint. Qwen: ragpu003. Llama: ragpu004.

Model	Fmt	tok/s	req/s	GPU-s/q	VRAM (GB)	C_{pass}
Qwen-7B	BF16	252.72 $\pm$ 0.56	0.074	13.44	54.98	\$0.0060
Qwen-7B	FP8	449.79 $\pm$ 97.53	0.120	8.64	54.99	\$0.0038
Qwen-7B	AWQ-4	418.15 $\pm$ 2.03	0.128	7.82	55.09	\$0.0035
Qwen-7B	GPTQ-4	488.26 $\pm$ 0.64	0.138	7.23	53.89	\$0.0032
Llama-8B	BF16	366.98 $\pm$ 1.90	0.099	10.13	55.94	\$0.0047
Llama-8B	FP8	481.42 $\pm$ 1.36	0.107	9.32	55.95	\$0.0043
Llama-8B	AWQ-4	391.11 $\pm$ 0.48	0.086	11.63	56.14	\$0.0056
Llama-8B	GPTQ-4	366.16 $\pm$ 0.79	0.079	12.60	56.01	\$0.0059

5 Discussion

Under this pinned A100 W8A16 stack, FP8-checkpoint serving tracks BF16 pass@1 closely on the evaluated distillations, but we do not establish ± 1 pp equivalence on MATH-500. Four-bit behavior is not a single story: Qwen keeps MATH-500 and GSM8K near BF16 while losing 5.6 pp on GPQA-Diamond under AWQ-4; Llama AWQ-4 is the MATH/GSM8K weak cell. Mean 4-bit lengthening on MATH-500 is modest and, when present, lives mostly on traces that also lose the answer. That is a sharper claim than “4-bit inflates reasoning” and a more honest one than “nothing happens.”

Pass@1 and stochastic output stability are related but distinct. Llama AWQ-4’s degradation appears in both mean accuracy and 5/5 consensus coverage. Qwen 4-bit formats retain relatively strong MATH-500 pass@1 while spending more tokens. No clear FP8–BF16 difference was detected in modal-agreement coverage or selective risk under these configurations; paired intervals that include zero do not prove equivalence. Repeated sampling sharply reduces selective risk on MATH-500 but costs about five generations per attempted item. The construction is not a general uncertainty estimator and is not a safety property; it is an abstention rule on MATH-500 that requires five generations before the decision.

We previously wrote that pinning vLLM 0.7.0 “removed” truncations and loops seen on vLLM 0.8.5. Those earlier runs also differed in harness, decoding flags (including a bug that dropped `repetition_penalty`), and prompt profile. The final grid still contains 25 loops. We therefore do not attribute pathology rates causally to the engine version. A clean stack $\times$ format factorial remains future work, not a result in this paper.

FP8 is a reasonable *accuracy* default on this A100 W8A16 path for these two distillations. On confirmation Condition B it is not the unique cheapest Qwen cell: Qwen GPTQ-4 has lower scenario C_{pass} (−45.9% vs. BF16) than Qwen FP8 (−36.0%), and the FP8 tok/s mean carries a 22% sample CV. It is still not Hopper native W8A8.

Cost is tokens and cost-per-token, measured here as GPU-seconds under a pricing scenario. Qwen 4-bit lengthening on the 56k MATH-500 grid is real; on the confirmation subset it does not, by itself, decide dollars. Batched 4-bit kernels can raise tok/s (Qwen GPTQ-4 +93.2% vs. Qwen BF16) or not (Llama GPTQ-4 −0.2% vs. Llama BF16). Single-request serving effects were architecture dependent: Qwen AWQ/GPTQ substantially exceeded BF16 throughput, whereas Llama AWQ/GPTQ did not. Peak allocated VRAM does not separate formats under a 0.75 utilization cap; do not read Table 8 as a memory-compression result.

6 Limitations

1. **A100 W8A16 fallback.** FP8 numbers do not transfer to Hopper native W8A8.
2. **Measured serving scope.** Throughput, latency, GPU-sec/query, and scenario C_{pass} are measured on balanced MATH-500 serving subsets (20 sequential prompts for Condition A; 100 batched prompts for Condition B), not the full 56k campaign. Pass@1 in C_{pass} is the 40-cell campaign mean. The \$1.50/A100-hour rate is a pricing scenario, not billed cluster cost. Confirmation Condition B pins `max_num_seqs=8`. Repetitions share sampling seed 20260816. Qwen and Llama confirmation jobs used different hosts (`ragpu003` vs. `ragpu004`); within-architecture comparisons are same-node. Qwen FP8 Condition B retains five wall-clock repeats with large timing variance at identical token counts. Peak allocated VRAM is the 0.75 engine pool, not isolated weight footprint. Compact confirmation JSON has no `finish_reason`. Energy is not measured.
3. **Pathology metrology.** The loop detector is identical-word runs ≥ 20 . Phrase-level cycles and `finish_reason` were not saved in the original campaign. Near-cap is a proxy.
4. **Modal agreement.** Five generations are required per attempted MATH problem. Only MATH-500 has five seeds in this campaign. Mathematical answer equivalence is evaluator- and task-specific (campaign LightEval 0.8.0 / math-verify). High agreement can still be wrong. The signal is not general uncertainty calibration, not softmax calibration, and not a safety property.
5. **Checkpoint provenance.** AWQ-4 is a community quantization of unknown calibration set and quantizer revision. GPTQ-4/FP8 are RedHatAI artifacts (appendix).
6. **Scope.** Two 7–8B DeepSeek-R1 distillations, English math/science, one pinned serving stack, one prompt. GSM8K/GPQA use three seeds.
7. **Source configs vs. launcher.** Files in `configs/models/` are not the QRM `inference.py` campaign. Effective KV-cache dtype is not stored per cell. Launcher flags such as `max_model_len=32768` are code/launcher-confirmed, not per-job runtime telemetry.

7 Conclusion

On a pinned vLLM 0.7.0 eager stack, FP8-checkpoint (W8A16) pass@1 stays close to BF16 for two R1 distillations, while 4-bit effects are model- and task-dependent and are visible to clustered pass@1 tests that maj@5 McNemar misses. Loops are rare but nonzero; near-cap terminations are more common for Qwen 4-bit on MATH-500. Mean token inflation, where it exists, is concentrated on failures. Observable 5-sample agreement on MATH-500 yields low selective error at strict consensus, at the cost of five generations, and Llama AWQ-4 reduces 5/5 coverage relative to BF16. Controlled batched serving makes Qwen GPTQ-4 the lowest scenario C_{pass} among Qwen cells (−45.9% vs. BF16 at \$1.50/A100-hour); Qwen FP8 is cheaper than Qwen BF16 (−36.0%) but with a large wall-clock spread; Llama AWQ/GPTQ raise C_{pass} . Single-request serving effects were architecture dependent, and allocated VRAM does not separate formats. The contribution is a stack-pinned measurement of those effects, not a new compressor and not a serving-stack shift.

Artifacts

Code, patches, validation JSON, and analysis reports: [Manish06N/reasoning-compression-lab](#). Per-cell records: `results/math500/`, `results/gsm8k/`, `results/gpqa/`. Canonical pass@1 tables: `revision_reanalysis_report.json`. Modal agreement: `modal_agreement_report.json`.

Measured serving (confirmation): `results/reports/measured_serving_confirmation/`. First unconstrained timing (provenance only): `results/measured_serving/`. See `paper/ARTIFACT.md`. GPQA is a gated benchmark; public records must not republish item text.

Acknowledgment

PARAM Rudra HPC at IIT Patna (National Supercomputing Mission, C-DAC) provided compute.

A Prompt

MATH-500 and the GSM8K/GPQA QRM templates use:

```
{question}
```

```
Please reason step by step, and put your final
answer within \boxed{}
```

with a `<think>` prefix on the official QRM path.

B Checkpoints

Cell	Hugging Face ID / revision
Qwen BF16	deepseek-ai/DeepSeek-R1-Distill-Qwen-7B 916b56a44061fd5cd7d6a8fb632557ed4f724f60
Qwen FP8	RedHatAI/DeepSeek-R1-Distill-Qwen-7B-FP8-dynamic ceb2fcd178d477cd21b92ffb43164c74165a212c
Qwen AWQ-4	jakiAJK/DeepSeek-R1-Distill-Qwen-7B_AWQ 1d082b3ded4e7cefb6b32cd5e07c7d409832816
Qwen GPTQ-4	RedHatAI/DeepSeek-R1-Distill-Qwen-7B-quantized.w4a16 9559a91ed463eecda64b93a532fe5aa8cc25f2eb
Llama BF16	deepseek-ai/DeepSeek-R1-Distill-Llama-8B 6a6f4aa4197940add57724a7707d069478df56b1
Llama FP8	RedHatAI/DeepSeek-R1-Distill-Llama-8B-FP8-dynamic 5d548d9169c53b7bb7ef0b3bc261509d5da6e3dd
Llama AWQ-4	jakiAJK/DeepSeek-R1-Distill-Llama-8B_AWQ 57360e84a0e7278c6b3abb453f2dc812b213badb
Llama GPTQ-4	RedHatAI/DeepSeek-R1-Distill-Llama-8B-quantized.w4a16 faa06ed0a4a5bdab76c4df12376e0c5228c0bdfe

AWQ-4 checkpoints are third-party community uploads; calibration corpus and quantization code revision are unknown. GPTQ-4 and FP8 are RedHatAI compressed-tensors artifacts. Dataset revision SHAs for MATH-500 / GSM8K / GPQA-Diamond were not pinned in the campaign launcher beyond the QRM task names.

Serving confirmation protocol and raw JSON live under `docs/MEASURED_SERVING_CONFIRMATION_PROTOCOL` and `results/measured_serving_confirmation/raw/`. Aggregates: `measured_serving_confirmation_repo`. An earlier unconstrained timing (`docs/MEASURED_SERVING_PROTOCOL.md`, `results/measured_serving/`) is provenance only and is not mixed with confirmation means.

C Reanalysis

Pathology, clustered bootstrap, and token strata are produced by the canonical reanalysis script from the released per-cell JSON. Loop threshold = 20. Near-cap threshold = 32,500. Bootstrap $B = 10,000$, seed 0.

Table 9: Paired coverage Δ versus matched BF16 (percentage points). Intervals that exclude zero are marked *. Selective-accuracy Δ intervals are in the machine-readable modal report.

Contrast	$\geq 3/5$ cov. Δ [95% CI]	$\geq 4/5$ cov. Δ [95% CI]	5/5 cov. Δ [95% CI]
Qwen FP8	+0.6 [−0.8, +2.0]	−0.8 [−2.2, +0.6]	+0.4 [−1.6, +2.4]
Qwen AWQ-4	−0.2 [−1.8, +1.2]	−2.2 [−4.0, −0.6]*	−2.0 [−4.4, +0.2]
Qwen GPTQ-4	−0.6 [−2.0, +0.8]	−0.6 [−2.2, +1.0]	−1.6 [−4.2, +0.8]
Llama FP8	−0.0 [−1.8, +1.8]	+0.8 [−1.6, +3.2]	+2.0 [−1.2, +5.2]
Llama AWQ-4	−0.8 [−2.8, +1.2]	−3.2 [−6.0, −0.6]*	−6.0 [−9.4, −2.6]*
Llama GPTQ-4	−0.2 [−2.2, +1.8]	−1.4 [−3.8, +1.0]	−0.8 [−4.2, +2.6]

D Modal-agreement details

The compact recovered artifact has 20,000 rows (SHA256 prefix `23e9ead021111959`; full digest is in the JSON sidecar). It stores extracted prediction representations, campaign `extractive_match`, and completion tokens. It does not store chain-of-thought or problem text. Campaign extraction used LightEval 0.8.0; a later MacBook LightEval 0.8.1 install was not used for these numbers. Campaign extractive match was reproduced on 20,000/20,000 rows before clustering. Equivalence clustering reported 0 symmetry violations, 0 transitivity violations, and 116 non-unique-mode ties among 4,000 groups. Bootstrap uses $B = 10,000$ problem resamples with seed 20260816. Secondary ECE/Brier scores (confidence = $m_i/5$) are in the JSON report and are not treated as a primary calibration result.

T_5 is the five-sample token-cost proxy: the sum of completion tokens over seeds 42–46. Mean / median / P90 T_5 are computed over all 500 items. Tokens per served query and tokens per correct served result use that same 500-item T_5 total in the numerator, so abstained queries are not free.

E Measured serving: single-stream, microbenchmark, and superseded first run

Condition A (Table 10) is sequential decode of a balanced 20-prompt MATH-500 subset (4 per difficulty level). Median latency is the mean of per-run medians. $R = 3$ except Llama AWQ-4 ($R = 5$; CV-triggered). Fixed-token microbenchmark (Table 11): 10 prompts $\times$ 512 tokens, $T = 0.0$, `ignore_eos`. Qwen microbenchmarks ran on `ragpu003`; Llama on `ragpu004`. Microbenchmark tok/s isolates kernel decode from trace-length variation and does not replace task-realistic serving.

The first unconstrained serving timing (`results/measured_serving/`) used unpinned `max_num_seqs`, an unbalanced Condition A first-20 draw, and mixed-node Llama repeats. Absolute tok/s from that run is not comparable to confirmation after pinning concurrency 8 and rebalancing the prompt mix. Those files remain as provenance; they are not averaged with confirmation and are not cited in the abstract.

Table 10: Confirmation single-stream serving (Condition A: 20 sequential prompts). Latency columns are means of per-run median/P90. C_{pass} uses campaign MATH-500 pass@1 and a \$1.50/A100-hour scenario. Tok/s is mean $\pm$ sample SD.

Model	Fmt	tok/s	Med. lat (s)	P90 lat (s)	GPU-s/q	C_{pass}
Qwen-7B	BF16	43.92 $\pm$ 0.04	59.19	244.83	102.51	\$0.0454
Qwen-7B	FP8	62.50 $\pm$ 0.08	43.60	197.01	84.60	\$0.0373
Qwen-7B	AWQ-4	76.89 $\pm$ 0.70	26.47	110.43	48.04	\$0.0215
Qwen-7B	GPTQ-4	82.67 $\pm$ 0.69	34.78	67.51	54.39	\$0.0242
Llama-8B	BF16	72.34 $\pm$ 0.27	32.40	126.10	66.43	\$0.0310
Llama-8B	FP8	78.75 $\pm$ 2.89	38.00	253.55	92.55	\$0.0431
Llama-8B	AWQ-4	70.20 $\pm$ 2.71	40.12	246.14	89.51	\$0.0431
Llama-8B	GPTQ-4	63.49 $\pm$ 0.41	46.31	242.83	86.95	\$0.0407

Table 11: Fixed-token decode microbenchmark (10 prompts $\times$ 512 tokens). Isolates kernel decode speed from trace-length variation. Secondary to task-realistic Conditions A/B.

Model	Fmt	Raw decode tok/s
Qwen-7B	BF16	205.41
Qwen-7B	FP8	437.19
Qwen-7B	AWQ-4	377.73
Qwen-7B	GPTQ-4	401.19
Llama-8B	BF16	362.01
Llama-8B	FP8	392.05
Llama-8B	AWQ-4	344.04
Llama-8B	GPTQ-4	302.54

References

- [1] Ekaterina Alimaskina, Darya Rudas, Denis Shveykin, Gleb Molodtsov, Pavel Vasiliev, and Aleksandr Beznosikov. Extreme low-bit inference in reasoning models: Failure modes and targeted recovery. *arXiv preprint arXiv:2606.02011*, 2026.
- [2] Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, et al. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.
- [3] Mehmet Hamza Erol, Batu El, Mirac Suzgun, Mert Yükeşgönül, and James Zou. Cost-of-pass: An economic framework for evaluating language models. *arXiv preprint arXiv:2504.13359*, 2025.
- [4] Elias Frantar, Saleh Ashkboos, Torsten Hoefer, and Dan Alistarh. GPTQ: Accurate post-training quantization for generative pre-trained transformers. *arXiv preprint arXiv:2210.17323*, 2022.
- [5] Elias Frantar, Roberto L. Castro, Jiale Chen, Torsten Hoefer, and Dan Alistarh. MARLIN: Mixed-precision auto-regressive parallel inference on large language models. *arXiv preprint arXiv:2408.11743*, 2024.
- [6] Yonatan Geifman and Ran El-Yaniv. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems*, 2017.

- [7] Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, Xiao Bi, et al. Deepseek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.
- [8] Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. Measuring mathematical problem solving with the MATH dataset. In *Neural Information Processing Systems (NeurIPS)*, 2021.
- [9] Andreas Hochlehnert, Hardik Bhatnagar, Vishaal Udandara, Samuel Albanie, Ameya Prabhu, and Matthias Bethge. A sober look at progress in language model reasoning: Pitfalls and paths to reproducibility. *arXiv preprint arXiv:2504.07086*, 2025.
- [10] Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. Efficient memory management for large language model serving with PagedAttention. In *Proceedings of the 29th Symposium on Operating Systems Principles (SOSP)*, 2023.
- [11] Zhen Li, Yupeng Su, Songmiao Wang, Runming Yang, Congkai Xie, Aofan Liu, Ming Li, Jiannong Cao, Yuan Xie, Ngai Wong, and Hongxia Yang. Quantization meets reasoning: Exploring and mitigating degradation of low-bit LLMs in mathematical reasoning. *arXiv preprint arXiv:2505.11574*, 2025.
- [12] Xinyu Lian, Walid Krichene, Beichen Huang, Masahiro Tanaka, Olatunji Ruwase, Li Zhang, and Minjia Zhang. Quantization inflates reasoning: Token inflation as a hidden cost of low-bit reasoning models. *arXiv preprint arXiv:2606.25519*, 2026.
- [13] Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let’s verify step by step. *arXiv preprint arXiv:2305.20050*, 2023.
- [14] Ji Lin, Jiaming Tang, Haotian Tang, Shang Yang, Wei-Ming Chen, Wei-Chen Wang, Guangxuan Xiao, Xingyu Dang, Chuang Gan, and Song Han. AWQ: Activation-aware weight quantization for LLM compression and acceleration. In *Proceedings of the Conference on Machine Learning and Systems (MLSys)*, 2024.
- [15] Ruikang Liu, Yuxuan Sun, Manyi Zhang, Haoli Bai, Xianzhi Yu, Tiezheng Yu, Chun Yuan, and Lu Hou. Quantization hurts reasoning? an empirical study on quantized reasoning models. *arXiv preprint arXiv:2504.04823*, 2025.
- [16] David Rein, Betty Li Hou, Asa Cooper Stickland, Jackson Petty, Richard Yuanzhe Pang, Julien Dirani, Julian Michael, and Samuel R Bowman. GPQA: A graduate-level google-proof Q&A benchmark. *arXiv preprint arXiv:2311.12022*, 2023.
- [17] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations (ICLR)*, 2023.
- [18] Guangxuan Xiao, Ji Lin, Mickael Seznec, Hao Wu, Julien Demouth, and Song Han. Smoothquant: Accurate and efficient post-training quantization for large language models. In *International Conference on Machine Learning (ICML)*, pages 38087–38099, 2023.
- [19] Mingyu Zhong, Guanchu Wang, Yu-Neng Chuang, and Na Zou. Quantized can still be calibrated: A unified framework to calibration in quantized large language models. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 30503–30517, 2025.

- [20] Thomas Zollo, Jimmy Wang, and Richard Zemel. Unsupervised confidence calibration for reasoning LLMs from a single generation. *arXiv preprint arXiv:2604.19444*, 2026.